KLAi

Your first Python: variables, lists and error messages, next to the spreadsheet

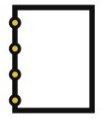
What does the same price-list task look like in Excel and in Python, and how do you read an error message when Python stops?

AI and Data Science Professional | Session 5 of 24 | Module B: Python for data

KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Run and save a notebook, and store values in variables with the right data type: int, float, str or bool

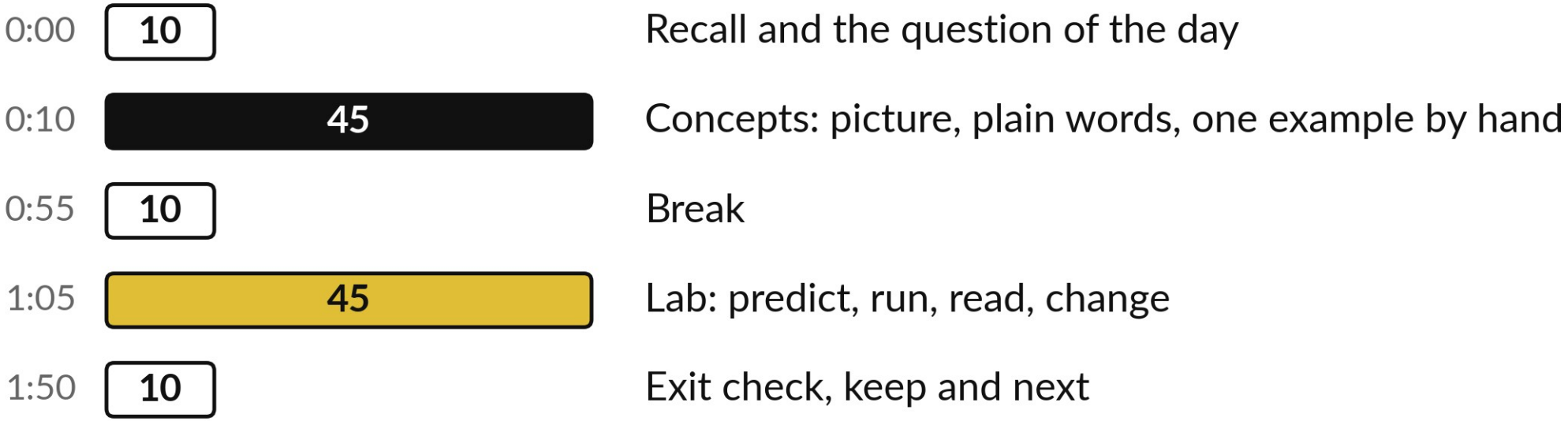


Hold many values in a list or a dictionary of key-value pairs, and look one up



Read a traceback from its last line, name the exception, and fix the typed-number-versus-string trap

Your two hours: concepts first, then a Python notebook lab



Pictures and plain words first. Then you open the notebook. Then the checks and what you keep.

Three questions from last time

1. You drew 200 samples of 30 orders and the 200 averages were not all the same. What is that variation called?

2. A survey asks only the customers who come to the till on a Saturday. What goes wrong?

3. Branch footfall and orders rise together. Does that show footfall causes orders?



Answer aloud first. Then we check together.

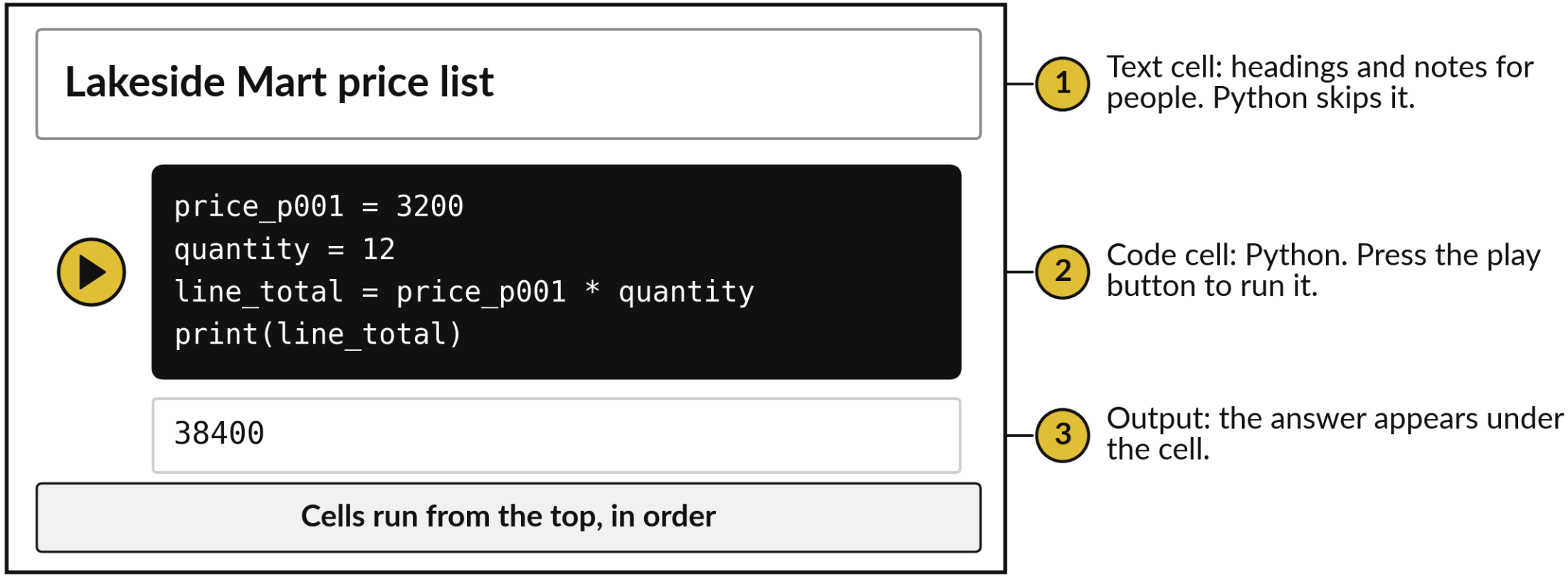
Answers to the three questions from Session 4

	Question	Short answer
	**You drew 200 samples of 30 orders and the 200 averages were not all the same. What is that variation called?**	Sampling variability. Its size is the standard error, and it gets smaller when each sample is bigger.
	**A survey asks only the customers who come to the till on a Saturday. What goes wrong?**	Selection bias. Shoppers who never come on a Saturday are left out, so the sample does not stand for all customers.
	**Branch footfall and orders rise together. Does that show footfall causes orders?**	No. Correlation is not causation. A confounder such as branch size can lift both.

Words we use today

	Word	What it means
	**Variable (assignment)**	A name that stands for a stored value, like a named cell in Excel. In Python you choose the name, such as price_p001. In Session 2 a variable was a column of a table. Here it is a name that holds a value.
	**Data type (int, float, str, bool)**	How Python stores a value: a whole number, a decimal number, text (a string) or True or False.
	**List**	Values in order inside square brackets. Each has an index, and the first index is 0. It is like one column of a sheet.
	**Dictionary (key-value pairs)**	Values stored under labels called keys. You look a value up by its key, as VLOOKUP does.
	**Exception (traceback)**	The error Python raises when a line cannot run. The traceback is its report. Its last line names the exception.

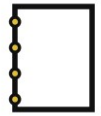
A notebook mixes text cells and code cells, and runs them from the top



A Jupyter notebook is a document of cells. Colab runs it in a browser. The kernel is its engine.

Run the cells from the top, and save a copy to Drive

Three habits keep a notebook honest.



Cells run in the order you press play. A later cell uses only what an earlier one stored.



If it goes odd, restart and run all cells from the top.



Save a copy in Drive. An idle session ends.

A variable is a name you give to a stored value, like a named cell in Excel

Excel or Sheets

	A	B
2	price_p001	3200
3	quantity	12
4	line_total	38400

Python

```
price_p001 = 3200

quantity = 12

line_total = price_p001 * quantity
```

`B4 holds =B2*B3 and shows 38400`

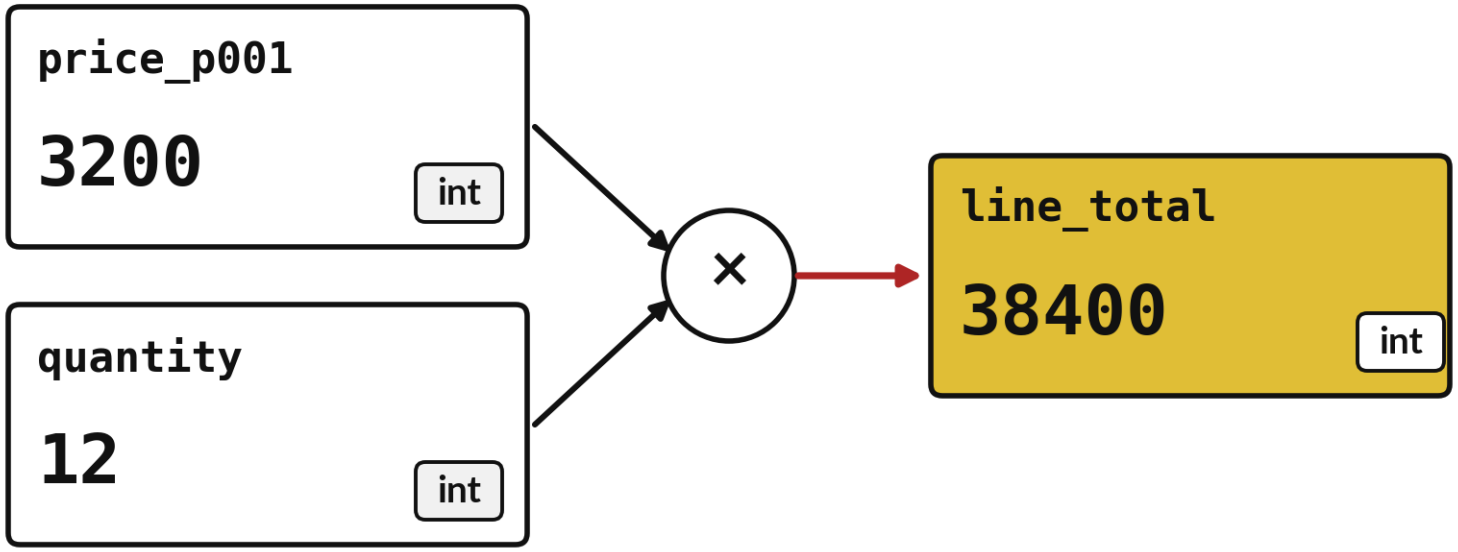
In Python the name you choose takes the place of the cell address.

Variable: a name that stands for a stored value.

Sheet: the value sits in cell B2, which you could name Price. Python: it is stored under the name price_p001.

In Session 2 a variable was a column of a table. In programming a variable is a name that holds a value.

Assignment works out the right side first, then stores the answer under the name




```
line_total = price_p001 * quantity
```

The right side is worked out first. The answer is then stored under the name on the left of the equals sign.

Assignment: the equals sign stores a value under a name.

Names use letters, digits and underscores. Case matters.

One equals sign stores. Two, ==, compare.

print() shows a value.

Every value has a data type: int, float, str or bool

`int`	`float`	`str`	`bool`
Whole number	Decimal number	String: text in quotes	True or False
Real example	Real example	Real example	Real example
`3200`	`6720.0`	`'Staples item 001'`	`False`
type() prints	type() prints	type() prints	type() prints
`<class 'int'>`	`<class 'float'>`	`<class 'str'>`	`<class 'bool'>`
Excel cousin	Excel cousin	Excel cousin	Excel cousin
A number with no decimals	A number with decimals	A text cell	TRUE or FALSE

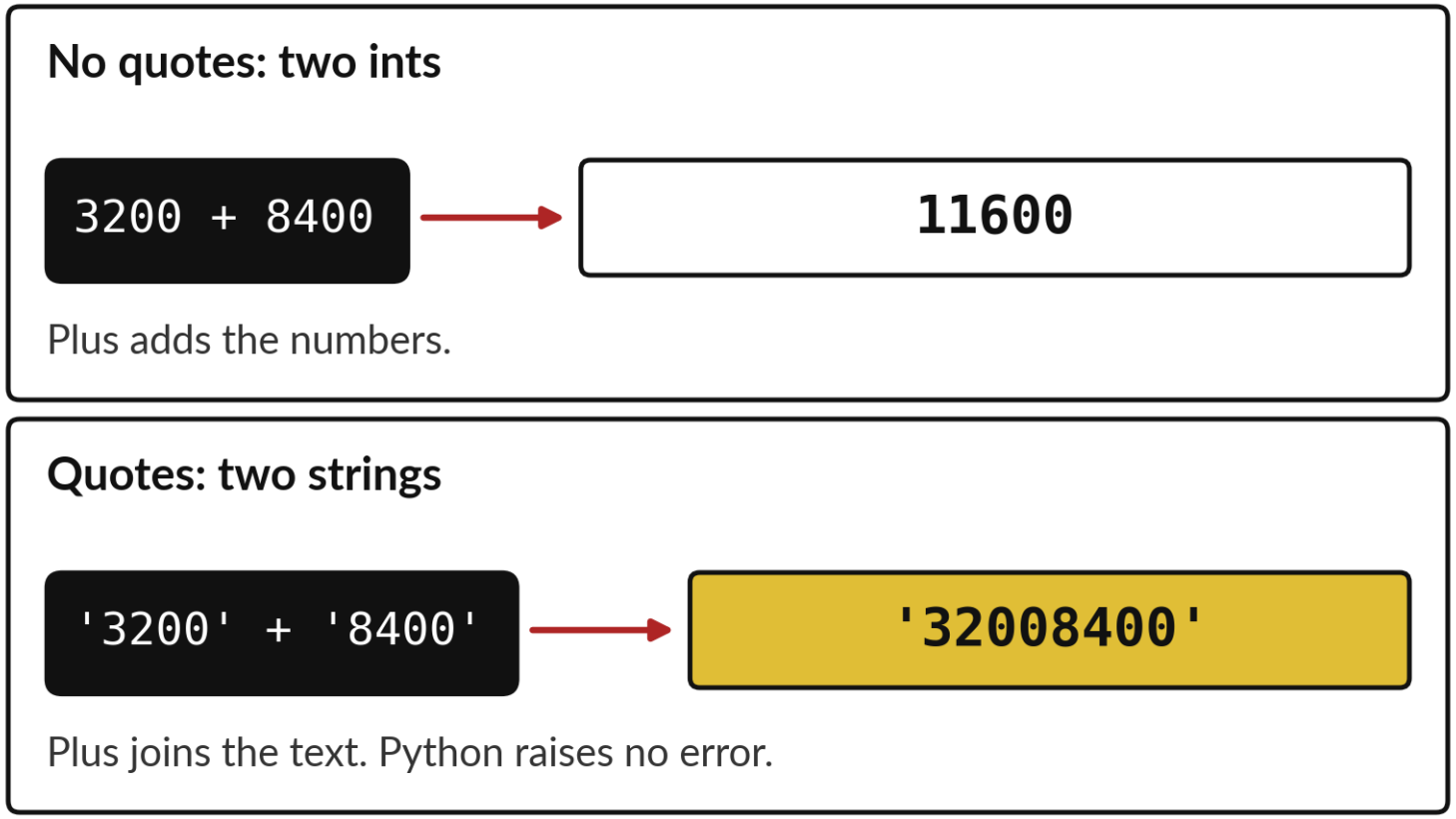
type() asks Python for a value's data type: how it is stored, as in Session 2.

Python does the arithmetic you know, plus a few new signs

	Python	Excel	What it does	Result
	**3200 * 12**	=3200*12	Multiply. Plus, minus and divide work as in Excel.	38400
	**sum(prices) / len(prices)**	=AVERAGE(range)	Divide. The answer is always a float.	6720.0
	**38400 // 1000**	=INT(38400/1000)	Floor division: the whole part only.	38
	**38400 % 1000**	=MOD(38400,1000)	Remainder after dividing.	400
	**2 ** 3**	=2^3	Power.	8

prices is the list of five prices from the next slides. Floor division and remainder come back in Session 6, when we turn spend into loyalty points.

Quotes turn a number into a string, and plus then joins instead of adding



A quiet wrong answer is worse than an error.

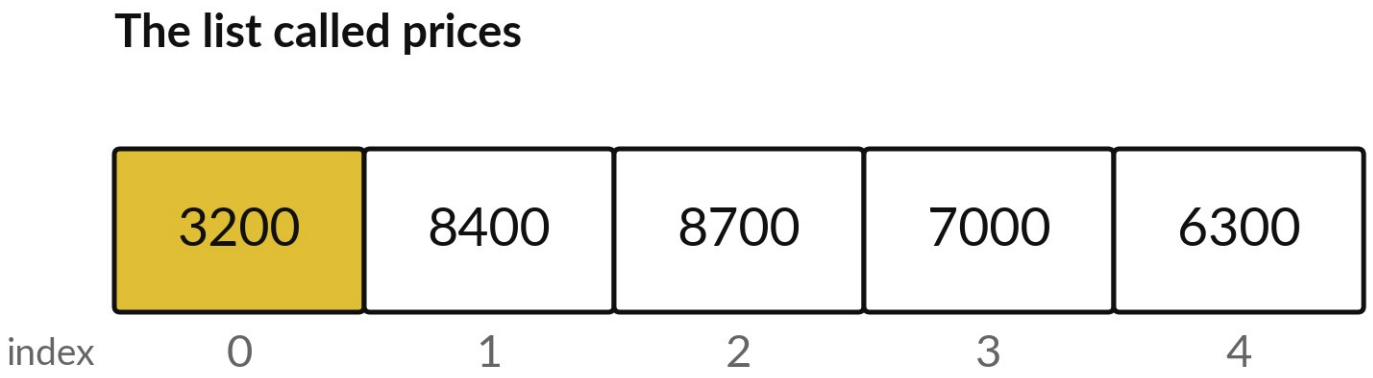
String (str): text inside quotes. A number in quotes is text.

Plus on ints adds. Plus on strings joins.

int() or float() converts a string: type conversion (casting).

Check type() when a total looks odd.

A list holds values in order, and the first position is index 0




```
prices[0]  ──→  3200   the first item
prices[4]  ──→  6300   the last of five items
len(prices) ──→  5     how many items
```

List: values in order, inside square brackets.

A list is like one column of a sheet.

Index: the position of an item, counting from 0.

len, sum and max work on the whole list.

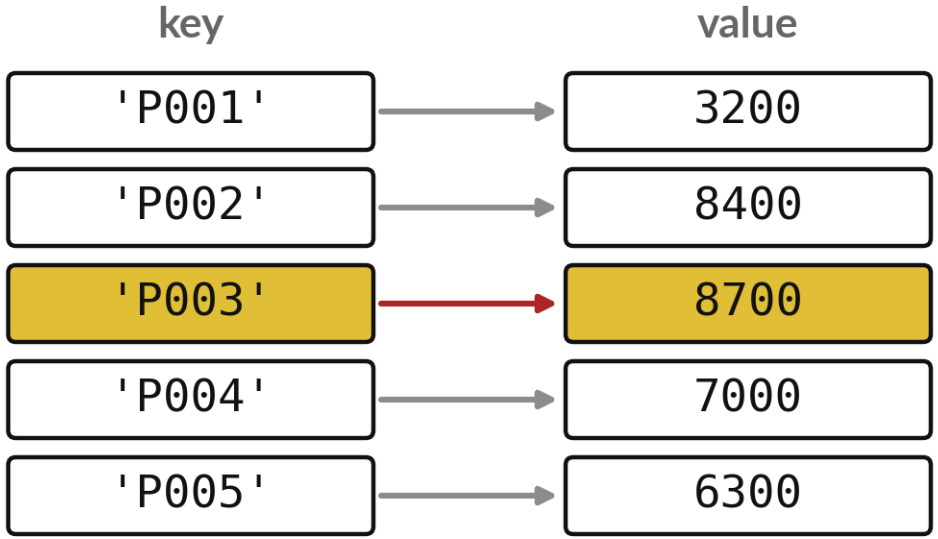
Functions on a list do what SUM, MAX and AVERAGE do on a column

	Python	Excel	What it does	Result
	**len(prices)**	=COUNT(E2:E6)	How many items	5
	**sum(prices)**	=SUM(E2:E6)	Add them all	33600
	**max(prices)**	=MAX(E2:E6)	The largest	8700
	**min(prices)**	=MIN(E2:E6)	The smallest	3200
	**sum(prices) / len(prices)**	=AVERAGE(E2:E6)	The mean	6720.0
	**prices.append(3600)**	Type the next price in E7	Add one item at the end	6 items

The five prices are in E2 to E6 of products.csv when you open it in a sheet. append changes the list itself.

A dictionary stores key-value pairs, so you find a value by its label

The dictionary called price_list




```
price_list['P003']  →  8700
```

Excel cousin: `=VLOOKUP("P003", table, 2, FALSE)`

Dictionary: values stored under labels called keys.

Each key is paired with one value: a key-value pair.

Curly brackets hold the pairs.

You ask by key, so the order does not matter.

List or dictionary: choose by how you will look things up

List

Values in order.

Find an item by its index: 0, 1, 2.

Excel twin: one column to add up.

Dictionary

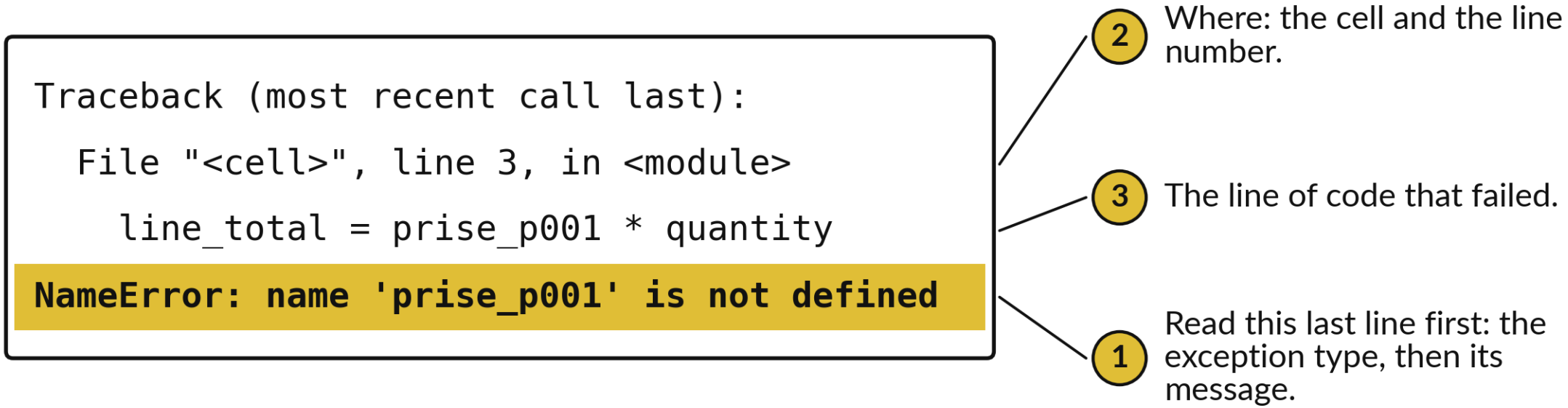
Values with labels.

Find an item by its key, such as 'P003'.

Excel twin: a VLOOKUP table, for a lookup.

Ask first: will I find it by position or by label?

A traceback is the report of a failure: read its last line first



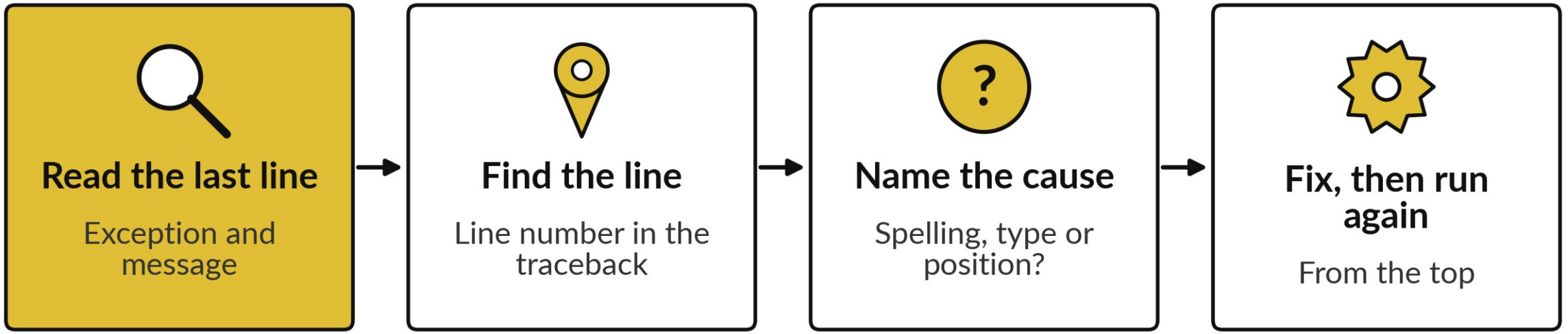
From a real run. Colab adds colours and an arrow. The last line is the same.

Five exceptions cover most beginner errors

	Exception	Last line says	What it means	Usual fix
	**NameError**	NameError: name 'prise_p001' is not defined	No variable has that name: a spelling slip, or its cell has not run.	Check the spelling. Run the earlier cell.
	**TypeError**	TypeError: unsupported operand type(s) for +: 'int' and 'str'	An operation met the wrong data type, such as int plus str.	Convert with int() or float().
	**KeyError**	KeyError: 'P099'	The dictionary has no such key.	Check the key spelling and its quotes.
	**IndexError**	IndexError: list index out of range	The list has no item at that index.	The last index is len minus 1.
	**SyntaxError**	SyntaxError: '(' was never closed	The line breaks a grammar rule.	Close every bracket and quote.

Messages copied from real runs in Python 3.10 or later. Newer Python versions may word a message slightly differently.

Four steps read any error and fix it

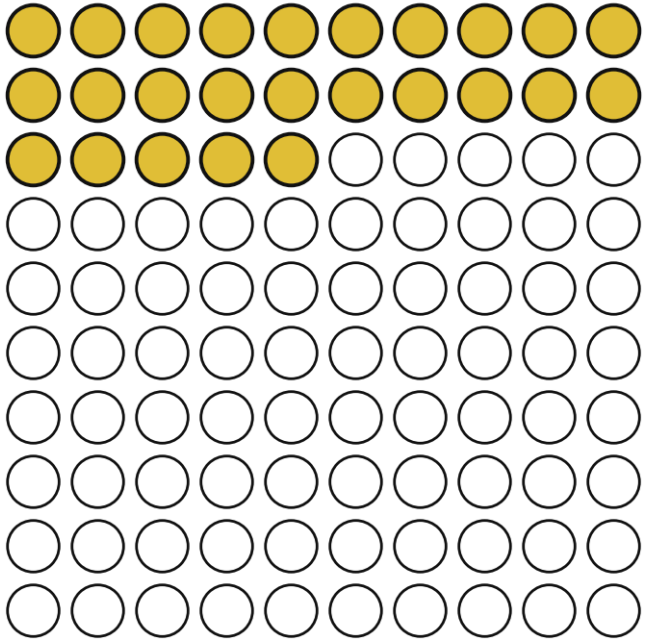


Change one thing at a time. An error is information.

A quarter of the raw export stores the price as text, so Python needs a conversion

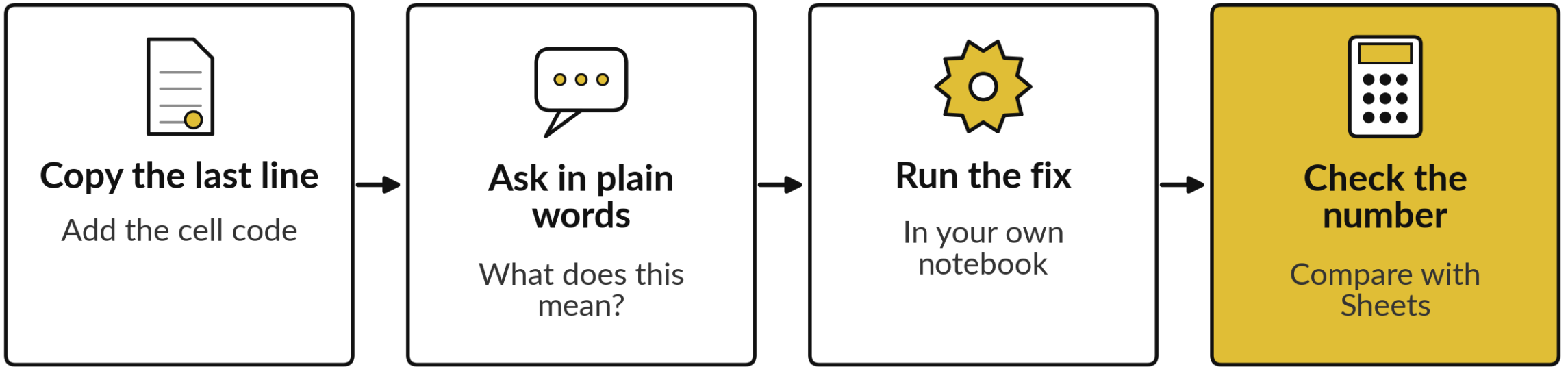
25%

10,345 of the 41,259 rows in orders_raw.csv hold unit_price as text such as UGX 7,800. Python cannot add text until you clean it.



raw rows in every 100 with a text price

A free chat assistant can explain an error, and you still check the fix



An assistant predicts likely words, so it can explain well and still be wrong about your data. Never paste personal or confidential data into a public chat tool.

Where you will use this at work: keep values in a list or a dictionary, and read the error

Your job	What it looks like	What today lets you do
Credit analyst at a bank, using Python and SQL	Loan amounts in a list. An interest rate per product in a dictionary.	Add up a list in one line. Look a rate up by key.
Finance officer at a SACCO, using Excel	Contributions copied from a statement as text, such as UGX 7,800.	Spot a string that looks like a number and convert it.
Data officer at a ministry, using pandas	A file of 150 products read into a list of prices.	Check the count, the largest value and the mean in three lines.
Analyst handing work to a colleague	A Colab notebook with notes in text cells.	Share a notebook that runs from the top.

Read the recipe before the lab: six lines read a real file into a list

```
import pandas as pd
products = pd.read_csv('products.csv')
all_prices = products['unit_price_ugx'].tolist()
print(len(all_prices))
print(max(all_prices))
print(round(sum(all_prices) / len(all_prices), 2))
```

```
150
381600
11523.33
```

In plain words

Line 1 loads pandas, a library for tables.

Line 2 reads products.csv into a table.

Line 3 puts the price column into a list.

Line 4: how many prices, 150.

Line 5: the largest, 381,600 UGX.

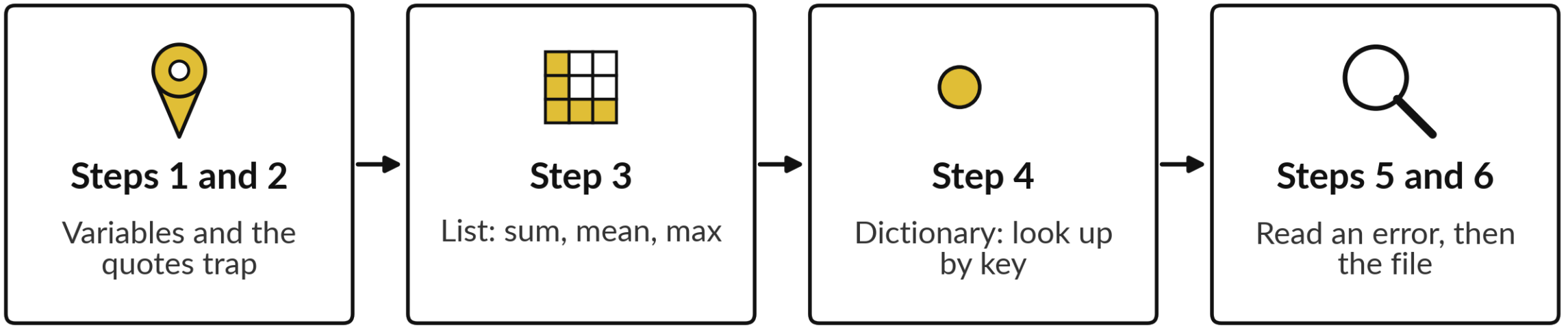
Line 6: the mean, 11523.33.



Part 2: the Python lab

Close the slides. Open the notebook.

The lab has six steps, and you keep a saved notebook at the end



For every task: predict, run, read the output, change one thing. None means no value yet: the lab cells start as None, and you replace it.

Check your numbers before you move on

	What you compute	You should get
	**line_total from price_p001 = 3200 and quantity = 12**	38,400
	**3200 + 8400 as strings, then as ints**	'32008400', then 11600
	**sum, mean and max of the five prices**	33,600, 6720.0 and 8,700
	**The sum after adding one more price of 3600**	37,200
	**The price of P003 in price_list, and its size after adding P006**	8,700, and 6
	**The exception name and the fixed basket total**	TypeError, and 20,300
	**Products in products.csv, the largest price, the mean**	150, 381,600 and 11523.33

If your number differs, find the task that went wrong. Do not change a number to match.

Five likely mistakes in the lab and how to fix them

	What you see	What it means	What to do
✗	**NameError: name 'price_p001' is not defined**	The cell that creates the name has not run, or the spelling differs.	Run the cells from the top. Match the spelling and the capital letters.
✗	**TypeError on a sum or a plus**	A string is mixed with a number.	Convert it with int() or float(). Check the quotes.
✗	**KeyError: 'P003'**	The key is missing from the dictionary, or it is not a string.	Write the key in quotes: 'P003'.
✗	**SyntaxError: was never closed**	A bracket or a quote was not closed.	Count the opening and closing signs on the line.
✓	**NOT YET after your answer is right**	The cell was run before the answer was typed, or the name differs.	Run the answer cell again, then the check line.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**Two prices are typed in quotes. What does '3200' + '8400' give, and why?**	'32008400'. Both values are strings, so plus joins them. Convert with int() to add.
	**A dictionary has no key P099. What does the last line of the error say, and what is the exception?**	KeyError: 'P099'. The exception is a KeyError.
	**How many items does a list of five prices have, and what is the index of the last one?**	Five items. The last index is 4, because counting starts at 0.

Save your notebook to Drive, with your error note

What you keep from today

Keep your saved notebook: the Lakeside price list built with variables, a list and a dictionary, a fixed number-versus-string trap, and a short note on one error you read and fixed.

Our question today

What does the same price-list task look like in Excel and in Python, and how do you read an error message when Python stops?

Next session

Session 6: Decisions, repeats and functions. Your code learns to choose with if, repeat with for, and reuse steps in a function.

Take away

Your lab file

A saved notebook

Your error note

Last line in plain words

Next session

Decisions and functions